Daniel Ernesto Mayo

Scientific Computing | HPC | Data & AI Infrastructure | Computational Chemistry

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Multidisciplinary professional combining a BSc in Chemistry (Physical Chemistry), Mechatronics training, Linux/HPC administration and Data Science/AI education. Experience spans scientific computing infrastructure, Slurm/JupyterHub environments, Python automation, computational chemistry concepts, data analysis and production systems. Particularly suited to roles where scientific workloads, infrastructure and AI/data engineering intersect.

SCIENTIFIC & TECHNICAL PROFILE

Scientific Computing HPC environments, Slurm, JupyterHub, Linux clusters and scientific workload support.	**Computational Chemistry** Physical Chemistry, DFT, molecular modeling, quantum dynamics and computational analysis.
Python & Data Python, SQL, scientific data processing, automation, Data Science and Machine Learning foundations.	**AI & MLOps** RAG, agentic AI concepts, MLflow/MLOps foundations and production-oriented AI infrastructure.
Industrial Perspective Mechatronics background, instrumentation, process analytics and infrastructure for technical environments.	**Infrastructure Foundation** Linux, virtualization, networking, Docker, Nginx, Proxmox and production operations.

RELEVANT EXPERIENCE

UNC - until 2016

Linux Server Administrator - HPC Infrastructure

Administered Linux scientific-computing infrastructure at Universidad Nacional de Córdoba.

- Supported research workloads on the Mendieta and Serafín HPC clusters.
- Worked with Slurm scheduling and JupyterHub collaborative computing environments.
- Connected infrastructure operations with real scientific and research workloads.

2024

Lecturer - Python Applied to Chemistry

Designed and delivered Python content for scientists and students.

- Focused on scientific data processing, automation and computational modeling workflows.

2008-2024

Infrastructure IT & Automation Consultant

Delivered systems, automation and data-processing support across technical and corporate environments.

- Built a strong operations foundation applicable to scientific platforms, laboratories and industrial data systems.

SELECTED SCIENTIFIC / DATA IMPACT

- HSK / Chinese language workstation: Streamlit + FFmpeg + faster-whisper + Argos Translate for local transcription and offline Chinese-Spanish translation.
- Multilingual RAG assistant: knowledge retrieval, Gemini, Docker and operational fallback flows.
- Scientific-computing operations: Linux HPC clusters, Slurm, JupyterHub and research-user support.

EDUCATION

BSc in Chemistry - Physical Chemistry - Universidad Nacional de Córdoba (2016)

Diploma in Data Science, Artificial Intelligence and Applications - Universidad Nacional de Córdoba (2024)

Higher Technician in Mechatronics - Universidad Tecnológica Nacional (2012)

Linux Server Administrator - Universidad Nacional de Córdoba (2014)

LANGUAGES & TARGET ROLES

Spanish: native | English: advanced B2/C1 | Portuguese: advanced | Mandarin Chinese: intermediate (HSK4)

Target roles: Scientific Computing Engineer, HPC Engineer, Data/AI Infrastructure Engineer, Computational Chemistry / Scientific Software, MLOps, Mining Technology, Laboratory Digitalization and Industrial Analytics.

SELECTED TECHNICAL STACK

Scientific compute Linux HPC, Slurm, JupyterHub, workload support	**Chemistry** Physical Chemistry, DFT, molecular modeling, quantum dynamics
Data Python, SQL, scientific data processing, ML foundations	**AI** RAG, agentic AI concepts, MLflow/MLOps foundations
Infrastructure Linux, virtualization, Docker, Nginx, Proxmox	**Industrial** Mechatronics, instrumentation, process analytics

PROFESSIONAL STRENGTHS

- Bridges scientific domain knowledge with infrastructure and software operations.
- Comfortable supporting researchers as well as production-oriented technical teams.
- Focus on reproducible computational workflows and operational reliability.
- Relevant cross-domain profile for HPC, mining, laboratories and industrial data platforms.

Recruiter note Open to remote and international opportunities. Portfolio and project case studies: https://shifudem.github.io/danielmayo/